

HelloScala.scala

```
object HelloScala {  
  def main(args: Array[String]): Unit = {  
    println("Hello Scala without Apache framework")  
  
    val nums = Array(1,2,3,4,5)  
    val sum = nums.sum  
  
    println("Sum = " + sum)  
  }  
}
```

Factorial.scala

```
object FactorialCalculator {  
  def factorial(n: Int): Int = {  
    if (n == 0)  
      1  
    else  
      n * factorial(n - 1)  
  }  
  def main(args: Array[String]): Unit = {  
    val number = 5  
    println("Factorial of " + number + " is " + factorial(number))  
  }  
}
```

SparkFactorial.scala

```
import org.apache.spark.sql.SparkSession  
  
object SparkFactorial {  
  def main(args: Array[String]): Unit = {  
  
    val spark = SparkSession.builder()  
      .appName("Factorial App")  
      .master("local[*]")  
      .getOrCreate()  
  
    val sc = spark.sparkContext  
  
    val num = 5  
    val result = sc.parallelize(1 to num).reduce(_ * _)  
  
    println("Factorial is: " + result)
```

```
    spark.stop()
  }
}
```

Without Apache Spark Framework

```
omkar@LAPTOP-S6G2S414:~$ wget https://downloads.lightbend.com/scala/2.13.14/scala-2.13.14.tgz
```

```
tar -xvzf scala-2.13.14.tgz
```

```
sudo mv scala-2.13.14 /opt/scala213
```

```
--2026-04-21 21:47:04-- https://downloads.lightbend.com/scala/2.13.14/scala-2.13.14.tgz
```

```
Resolving downloads.lightbend.com (downloads.lightbend.com)... 18.66.41.115,
```

```
18.66.41.64, 18.66.41.108, ...
```

```
Connecting to downloads.lightbend.com (downloads.lightbend.com)|18.66.41.115|:443...
connected.
```

```
HTTP request sent, awaiting response... 200 OK
```

```
Length: 24437520 (23M) [application/octet-stream]
```

```
Saving to: 'scala-2.13.14.tgz'
```

```
scala-2.13.14.tgz      100%[=====>]
23.30M  8.73MB/s   in 2.7s
```

```
2026-04-21 21:47:08 (8.73 MB/s) - 'scala-2.13.14.tgz' saved [24437520/24437520]
```

```
scala-2.13.14/
```

```
scala-2.13.14/bin/
```

```
scala-2.13.14/bin/scalac.bat
```

```
scala-2.13.14/bin/scaladoc
```

```
scala-2.13.14/bin/scalac
```

```
scala-2.13.14/bin/scalap.bat
```

```
scala-2.13.14/bin/scalap
```

```
scala-2.13.14/bin/scala.bat
```

```
scala-2.13.14/bin/scaladoc.bat
```

```
scala-2.13.14/bin/scala
```

```
scala-2.13.14/bin/fsc.bat
```

```
scala-2.13.14/bin/fsc
```

```
scala-2.13.14/NOTICE
```

```
scala-2.13.14/man/
```

```
scala-2.13.14/man/man1/
```

```
scala-2.13.14/man/man1/scaladoc.1
```

```
scala-2.13.14/man/man1/fsc.1
```

```
scala-2.13.14/man/man1/scalap.1
```

```
scala-2.13.14/man/man1/scala.1
```

```
scala-2.13.14/man/man1/scalac.1
```

```
scala-2.13.14/lib/
```

scala-2.13.14/lib/scala-library.jar
scala-2.13.14/lib/java-diff-utils-4.12.jar
scala-2.13.14/lib/scala-reflect.jar
scala-2.13.14/lib/scalap-2.13.14.jar
scala-2.13.14/lib/scala-compiler.jar
scala-2.13.14/lib/jline-3.25.1.jar
scala-2.13.14/lib/jna-5.14.0.jar
scala-2.13.14/doc/
scala-2.13.14/doc/LICENSE.md
scala-2.13.14/doc/tools/
scala-2.13.14/doc/tools/scalap.html
scala-2.13.14/doc/tools/scaladoc.html
scala-2.13.14/doc/tools/images/
scala-2.13.14/doc/tools/images/scala_logo.png
scala-2.13.14/doc/tools/images/external.gif
scala-2.13.14/doc/tools/scalac.html
scala-2.13.14/doc/tools/index.html
scala-2.13.14/doc/tools/scala.html
scala-2.13.14/doc/tools/css/
scala-2.13.14/doc/tools/css/style.css
scala-2.13.14/doc/tools/fsc.html
scala-2.13.14/doc/License.rtf
scala-2.13.14/doc/internal/
scala-2.13.14/doc/internal/tastyreader.md
scala-2.13.14/doc/licenses/
scala-2.13.14/doc/licenses/mit_jquery.txt
scala-2.13.14/doc/licenses/bsd_jline.txt
scala-2.13.14/doc/licenses/bsd_asm.txt
scala-2.13.14/doc/licenses/apache_jna.txt
scala-2.13.14/doc/README
scala-2.13.14/LICENSE

omkar@LAPTOP-S6G2S414:~\$ scala -version

Scala code runner version 2.13.14 -- Copyright 2002-2024, LAMP/EPFL and Lightbend, Inc.

omkar@LAPTOP-S6G2S414:~/scala-basic\$ scalac HelloScala.scala

omkar@LAPTOP-S6G2S414:~/scala-basic\$ scala HelloScala

Hello Scala without Apache framework

Sum = 15

omkar@LAPTOP-S6G2S414:~/scala-basic\$ scalac Factorial.scala

omkar@LAPTOP-S6G2S414:~/scala-basic\$ scala FactorialCalculator

Factorial of 5 is 120

With Apache Spark Framework

```
adding: SparkFactorial$.class(in = 3520) (out= 1617)(deflated 54%)
```

adding: SparkFactorial.class(in = 731) (out= 563)(deflated 22%)

omkar@LAPTOP-S6G2S414:~/scala-basic\$ spark-submit --class SparkFactorial
SparkFactorial.jar

WARNING: Using incubator modules: jdk.incubator.vector

Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties

26/04/21 21:50:19 WARN Utils: Your hostname, LAPTOP-S6G2S414, resolves to a loopback
address: 127.0.1.1; using 10.255.255.254 instead (on interface lo)

26/04/21 21:50:19 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address

Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties

26/04/21 21:50:22 INFO SparkContext: Running Spark version 4.1.1

26/04/21 21:50:22 INFO SparkContext: OS info Linux, 5.15.167.4-microsoft-standard-WSL2,
amd64

26/04/21 21:50:22 INFO SparkContext: Java version 17.0.18+8-Ubuntu-124.04.1

26/04/21 21:50:22 WARN NativeCodeLoader: Unable to load native-hadoop library for your
platform... using builtin-java classes where applicable

26/04/21 21:50:22 INFO ResourceUtils:

=====

26/04/21 21:50:22 INFO ResourceUtils: No custom resources configured for spark.driver.

26/04/21 21:50:22 INFO ResourceUtils:

=====

26/04/21 21:50:22 INFO SparkContext: Submitted application: Factorial App

26/04/21 21:50:22 INFO SecurityManager: Changing view acls to: omkar

26/04/21 21:50:22 INFO SecurityManager: Changing modify acls to: omkar

26/04/21 21:50:22 INFO SecurityManager: Changing view acls groups to: omkar

26/04/21 21:50:22 INFO SecurityManager: Changing modify acls groups to: omkar

26/04/21 21:50:22 INFO SecurityManager: SecurityManager: authentication disabled; ui acls
disabled; users with view permissions: omkar groups with view permissions: EMPTY; users
with modify permissions: omkar; groups with modify permissions: EMPTY; RPC SSL disabled

26/04/21 21:50:23 INFO Utils: Successfully started service 'sparkDriver' on port 44833.

26/04/21 21:50:23 INFO SparkEnv: Registering MapOutputTracker

26/04/21 21:50:23 INFO SparkEnv: Registering BlockManagerMaster

26/04/21 21:50:23 INFO BlockManagerMasterEndpoint: Using

org.apache.spark.storage.DefaultTopologyMapper for getting topology information

26/04/21 21:50:23 INFO BlockManagerMasterEndpoint: BlockManagerMasterEndpoint up

26/04/21 21:50:23 INFO SparkEnv: Registering BlockManagerMasterHeartbeat

26/04/21 21:50:23 INFO DiskBlockManager: Created local directory at /tmp/blockmgr-
3c8a1cd1-1504-4d4b-aaa9-8dbf270f9070

26/04/21 21:50:23 INFO SparkEnv: Registering OutputCommitCoordinator

26/04/21 21:50:23 INFO JettyUtils: Start Jetty 0.0.0.0:4040 for SparkUI

26/04/21 21:50:23 INFO Utils: Successfully started service 'SparkUI' on port 4040.

26/04/21 21:50:24 INFO SparkContext: Added JAR file:/home/omkar/scala-
basic/SparkFactorial.jar at spark://10.255.255.254:44833/jars/SparkFactorial.jar with
timestamp 1776808222293

26/04/21 21:50:24 INFO ResourceProfile: Default ResourceProfile created, executor
resources: Map(cores -> name: cores, amount: 1, script: , vendor: , memory -> name:

memory, amount: 1024, script: , vendor: , offHeap -> name: offHeap, amount: 0, script: , vendor:), task resources: Map(cpus -> name: cpus, amount: 1.0)

26/04/21 21:50:24 INFO ResourceProfile: Limiting resource is cpu

26/04/21 21:50:24 INFO ResourceProfileManager: Added ResourceProfile id: 0

26/04/21 21:50:24 INFO SecurityManager: Changing view acls to: omkar

26/04/21 21:50:24 INFO SecurityManager: Changing modify acls to: omkar

26/04/21 21:50:24 INFO SecurityManager: Changing view acls groups to: omkar

26/04/21 21:50:24 INFO SecurityManager: Changing modify acls groups to: omkar

26/04/21 21:50:24 INFO SecurityManager: SecurityManager: authentication disabled; ui acls disabled; users with view permissions: omkar groups with view permissions: EMPTY; users with modify permissions: omkar; groups with modify permissions: EMPTY; RPC SSL disabled

26/04/21 21:50:24 INFO Executor: Starting executor ID driver on host 10.255.255.254

26/04/21 21:50:24 INFO Executor: OS info Linux, 5.15.167.4-microsoft-standard-WSL2, amd64

26/04/21 21:50:24 INFO Executor: Java version 17.0.18+8-Ubuntu-124.04.1

26/04/21 21:50:24 INFO Executor: Starting executor with user classpath (userClassPathFirst = false): "

26/04/21 21:50:24 INFO Executor: Created or updated repl class loader org.apache.spark.util.MutableURLClassLoader@1ccdc7aa for default.

26/04/21 21:50:24 INFO Executor: Fetching spark://10.255.255.254:44833/jars/SparkFactorial.jar with timestamp 1776808222293

26/04/21 21:50:24 INFO TransportClientFactory: Successfully created connection to /10.255.255.254:44833 after 38 ms (0 ms spent in bootstraps)

26/04/21 21:50:24 INFO Utils: Fetching spark://10.255.255.254:44833/jars/SparkFactorial.jar to /tmp/spark-914fcb19-43a4-461f-890e-530d24f65072/userFiles-377de2a6-5265-43f8-a447-ed31cae3a3f2/fetchFileTemp4509670053665637893.tmp

26/04/21 21:50:24 INFO Executor: Adding file:/tmp/spark-914fcb19-43a4-461f-890e-530d24f65072/userFiles-377de2a6-5265-43f8-a447-ed31cae3a3f2/SparkFactorial.jar to class loader default

26/04/21 21:50:24 INFO Utils: Successfully started service 'org.apache.spark.network.netty.NettyBlockTransferService' on port 46575.

26/04/21 21:50:24 INFO NettyBlockTransferService: Server created on 10.255.255.254:46575

26/04/21 21:50:24 INFO BlockManager: Using org.apache.spark.storage.RandomBlockReplicationPolicy for block replication policy

26/04/21 21:50:24 INFO BlockManagerMaster: Registering BlockManager BlockManagerId(driver, 10.255.255.254, 46575, None)

26/04/21 21:50:24 INFO BlockManagerMasterEndpoint: Registering block manager 10.255.255.254:46575 with 434.4 MiB RAM, BlockManagerId(driver, 10.255.255.254, 46575, None)

26/04/21 21:50:24 INFO BlockManagerMaster: Registered BlockManager BlockManagerId(driver, 10.255.255.254, 46575, None)

26/04/21 21:50:24 INFO BlockManager: Initialized BlockManager: BlockManagerId(driver, 10.255.255.254, 46575, None)

26/04/21 21:50:27 INFO SparkContext: Starting job: reduce at SparkFactorial.scala:14

26/04/21 21:50:27 INFO DAGScheduler: Got job 0 (reduce at SparkFactorial.scala:14) with 16 output partitions

26/04/21 21:50:27 INFO DAGScheduler: Final stage: ResultStage 0 (reduce at SparkFactorial.scala:14)

26/04/21 21:50:27 INFO DAGScheduler: Parents of final stage: List()

26/04/21 21:50:27 INFO DAGScheduler: Missing parents: List()

26/04/21 21:50:27 INFO DAGScheduler: Missing parents found for ResultStage 0: List()

26/04/21 21:50:27 INFO DAGScheduler: Submitting ResultStage 0 (ParallelCollectionRDD[0] at parallelize at SparkFactorial.scala:14), which has no missing parents

26/04/21 21:50:27 INFO MemoryStore: MemoryStore started with capacity 434.4 MiB

26/04/21 21:50:27 INFO MemoryStore: Block broadcast_0 stored as values in memory (estimated size 3.0 KiB, free 434.4 MiB)

26/04/21 21:50:27 INFO MemoryStore: Block broadcast_0_piece0 stored as bytes in memory (estimated size 1810.0 B, free 434.4 MiB)

26/04/21 21:50:27 INFO SparkContext: Created broadcast 0 from broadcast at DAGScheduler.scala:1686

26/04/21 21:50:27 INFO DAGScheduler: Submitting 16 missing tasks from ResultStage 0 (ParallelCollectionRDD[0] at parallelize at SparkFactorial.scala:14) (first 15 tasks are for partitions Vector(0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14))

26/04/21 21:50:27 INFO TaskSchedulerImpl: Adding task set 0.0 with 16 tasks resource profile 0

26/04/21 21:50:27 INFO TaskSetManager: Starting task 0.0 in stage 0.0 (TID 0) (10.255.255.254,executor driver, partition 0, PROCESS_LOCAL, 9945 bytes)

26/04/21 21:50:27 INFO TaskSetManager: Starting task 1.0 in stage 0.0 (TID 1) (10.255.255.254,executor driver, partition 1, PROCESS_LOCAL, 9945 bytes)

26/04/21 21:50:27 INFO TaskSetManager: Starting task 2.0 in stage 0.0 (TID 2) (10.255.255.254,executor driver, partition 2, PROCESS_LOCAL, 9945 bytes)

26/04/21 21:50:27 INFO TaskSetManager: Starting task 3.0 in stage 0.0 (TID 3) (10.255.255.254,executor driver, partition 3, PROCESS_LOCAL, 9945 bytes)

26/04/21 21:50:27 INFO TaskSetManager: Starting task 4.0 in stage 0.0 (TID 4) (10.255.255.254,executor driver, partition 4, PROCESS_LOCAL, 9945 bytes)

26/04/21 21:50:27 INFO TaskSetManager: Starting task 5.0 in stage 0.0 (TID 5) (10.255.255.254,executor driver, partition 5, PROCESS_LOCAL, 9945 bytes)

26/04/21 21:50:27 INFO TaskSetManager: Starting task 6.0 in stage 0.0 (TID 6) (10.255.255.254,executor driver, partition 6, PROCESS_LOCAL, 9945 bytes)

26/04/21 21:50:27 INFO TaskSetManager: Starting task 7.0 in stage 0.0 (TID 7) (10.255.255.254,executor driver, partition 7, PROCESS_LOCAL, 9945 bytes)

26/04/21 21:50:27 INFO TaskSetManager: Starting task 8.0 in stage 0.0 (TID 8) (10.255.255.254,executor driver, partition 8, PROCESS_LOCAL, 9945 bytes)

26/04/21 21:50:27 INFO TaskSetManager: Starting task 9.0 in stage 0.0 (TID 9) (10.255.255.254,executor driver, partition 9, PROCESS_LOCAL, 9945 bytes)

26/04/21 21:50:27 INFO TaskSetManager: Starting task 10.0 in stage 0.0 (TID 10) (10.255.255.254,executor driver, partition 10, PROCESS_LOCAL, 9945 bytes)

26/04/21 21:50:27 INFO TaskSetManager: Starting task 11.0 in stage 0.0 (TID 11) (10.255.255.254,executor driver, partition 11, PROCESS_LOCAL, 9945 bytes)

26/04/21 21:50:27 INFO TaskSetManager: Starting task 12.0 in stage 0.0 (TID 12) (10.255.255.254,executor driver, partition 12, PROCESS_LOCAL, 9945 bytes)

26/04/21 21:50:27 INFO TaskSetManager: Starting task 13.0 in stage 0.0 (TID 13)
(10.255.255.254,executor driver, partition 13, PROCESS_LOCAL, 9945 bytes)
26/04/21 21:50:27 INFO TaskSetManager: Starting task 14.0 in stage 0.0 (TID 14)
(10.255.255.254,executor driver, partition 14, PROCESS_LOCAL, 9945 bytes)
26/04/21 21:50:27 INFO TaskSetManager: Starting task 15.0 in stage 0.0 (TID 15)
(10.255.255.254,executor driver, partition 15, PROCESS_LOCAL, 9945 bytes)
26/04/21 21:50:27 INFO Executor: Running task 0.0 in stage 0.0 (TID 0)
26/04/21 21:50:27 INFO Executor: Running task 7.0 in stage 0.0 (TID 7)
26/04/21 21:50:27 INFO Executor: Running task 5.0 in stage 0.0 (TID 5)
26/04/21 21:50:27 INFO Executor: Running task 4.0 in stage 0.0 (TID 4)
26/04/21 21:50:27 INFO Executor: Running task 1.0 in stage 0.0 (TID 1)
26/04/21 21:50:27 INFO Executor: Running task 8.0 in stage 0.0 (TID 8)
26/04/21 21:50:27 INFO Executor: Running task 3.0 in stage 0.0 (TID 3)
26/04/21 21:50:27 INFO Executor: Running task 2.0 in stage 0.0 (TID 2)
26/04/21 21:50:27 INFO Executor: Running task 6.0 in stage 0.0 (TID 6)
26/04/21 21:50:27 INFO Executor: Running task 9.0 in stage 0.0 (TID 9)
26/04/21 21:50:27 INFO Executor: Running task 10.0 in stage 0.0 (TID 10)
26/04/21 21:50:27 INFO Executor: Running task 11.0 in stage 0.0 (TID 11)
26/04/21 21:50:27 INFO Executor: Running task 12.0 in stage 0.0 (TID 12)
26/04/21 21:50:27 INFO Executor: Running task 13.0 in stage 0.0 (TID 13)
26/04/21 21:50:27 INFO Executor: Running task 15.0 in stage 0.0 (TID 15)
26/04/21 21:50:27 INFO Executor: Running task 14.0 in stage 0.0 (TID 14)
26/04/21 21:50:27 INFO Executor: Finished task 0.0 in stage 0.0 (TID 0). 1144 bytes result
sent to driver
26/04/21 21:50:27 INFO Executor: Finished task 3.0 in stage 0.0 (TID 3). 1160 bytes result
sent to driver
26/04/21 21:50:27 INFO Executor: Finished task 11.0 in stage 0.0 (TID 11). 1144 bytes result
sent to driver
26/04/21 21:50:27 INFO Executor: Finished task 15.0 in stage 0.0 (TID 15). 1160 bytes result
sent to driver
26/04/21 21:50:27 INFO Executor: Finished task 9.0 in stage 0.0 (TID 9). 1160 bytes result
sent to driver
26/04/21 21:50:27 INFO Executor: Finished task 1.0 in stage 0.0 (TID 1). 1144 bytes result
sent to driver
26/04/21 21:50:27 INFO Executor: Finished task 6.0 in stage 0.0 (TID 6). 1160 bytes result
sent to driver
26/04/21 21:50:27 INFO Executor: Finished task 8.0 in stage 0.0 (TID 8). 1144 bytes result
sent to driver
26/04/21 21:50:27 INFO Executor: Finished task 10.0 in stage 0.0 (TID 10). 1144 bytes result
sent to driver
26/04/21 21:50:27 INFO Executor: Finished task 5.0 in stage 0.0 (TID 5). 1144 bytes result
sent to driver
26/04/21 21:50:27 INFO Executor: Finished task 7.0 in stage 0.0 (TID 7). 1144 bytes result
sent to driver
26/04/21 21:50:27 INFO Executor: Finished task 13.0 in stage 0.0 (TID 13). 1144 bytes result
sent to driver

26/04/21 21:50:27 INFO Executor: Finished task 4.0 in stage 0.0 (TID 4). 1144 bytes result sent to driver

26/04/21 21:50:27 INFO Executor: Finished task 14.0 in stage 0.0 (TID 14). 1144 bytes result sent to driver

26/04/21 21:50:27 INFO Executor: Finished task 2.0 in stage 0.0 (TID 2). 1144 bytes result sent to driver

26/04/21 21:50:27 INFO Executor: Finished task 12.0 in stage 0.0 (TID 12). 1160 bytes result sent to driver

26/04/21 21:50:27 INFO TaskSetManager: Finished task 13.0 in stage 0.0 (TID 13) in 352 ms on 10.255.255.254 (executor driver) (1/16)

26/04/21 21:50:28 INFO TaskSetManager: Finished task 2.0 in stage 0.0 (TID 2) in 381 ms on 10.255.255.254 (executor driver) (2/16)

26/04/21 21:50:28 INFO TaskSetManager: Finished task 8.0 in stage 0.0 (TID 8) in 374 ms on 10.255.255.254 (executor driver) (3/16)

26/04/21 21:50:28 INFO TaskSetManager: Finished task 5.0 in stage 0.0 (TID 5) in 380 ms on 10.255.255.254 (executor driver) (4/16)

26/04/21 21:50:28 INFO TaskSetManager: Finished task 10.0 in stage 0.0 (TID 10) in 373 ms on 10.255.255.254 (executor driver) (5/16)

26/04/21 21:50:28 INFO TaskSetManager: Finished task 9.0 in stage 0.0 (TID 9) in 378 ms on 10.255.255.254 (executor driver) (6/16)

26/04/21 21:50:28 INFO TaskSetManager: Finished task 6.0 in stage 0.0 (TID 6) in 384 ms on 10.255.255.254 (executor driver) (7/16)

26/04/21 21:50:28 INFO TaskSetManager: Finished task 1.0 in stage 0.0 (TID 1) in 398 ms on 10.255.255.254 (executor driver) (8/16)

26/04/21 21:50:28 INFO TaskSetManager: Finished task 7.0 in stage 0.0 (TID 7) in 389 ms on 10.255.255.254 (executor driver) (9/16)

26/04/21 21:50:28 INFO TaskSetManager: Finished task 3.0 in stage 0.0 (TID 3) in 396 ms on 10.255.255.254 (executor driver) (10/16)

26/04/21 21:50:28 INFO TaskSetManager: Finished task 12.0 in stage 0.0 (TID 12) in 377 ms on 10.255.255.254 (executor driver) (11/16)

26/04/21 21:50:28 INFO TaskSetManager: Finished task 11.0 in stage 0.0 (TID 11) in 382 ms on 10.255.255.254 (executor driver) (12/16)

26/04/21 21:50:28 INFO TaskSetManager: Finished task 14.0 in stage 0.0 (TID 14) in 378 ms on 10.255.255.254 (executor driver) (13/16)

26/04/21 21:50:28 INFO TaskSetManager: Finished task 0.0 in stage 0.0 (TID 0) in 440 ms on 10.255.255.254 (executor driver) (14/16)

26/04/21 21:50:28 INFO TaskSetManager: Finished task 4.0 in stage 0.0 (TID 4) in 407 ms on 10.255.255.254 (executor driver) (15/16)

26/04/21 21:50:28 INFO TaskSetManager: Finished task 15.0 in stage 0.0 (TID 15) in 382 ms on 10.255.255.254 (executor driver) (16/16)

26/04/21 21:50:28 INFO DAGScheduler: ResultStage 0 (reduce at SparkFactorial.scala:14) finished in 656 ms

26/04/21 21:50:28 INFO TaskSchedulerImpl: Removed TaskSet 0.0 whose tasks have all completed, from pool

26/04/21 21:50:28 INFO DAGScheduler: Job 0 is finished. Cancelling potential speculative or zombie tasks for this job

26/04/21 21:50:28 INFO TaskSchedulerImpl: Canceling stage 0

26/04/21 21:50:28 INFO TaskSchedulerImpl: Killing all running tasks in stage 0: Stage finished
26/04/21 21:50:28 INFO DAGScheduler: Job 0 finished: reduce at SparkFactorial.scala:14, took 738.337263 ms
Factorial is: 120
26/04/21 21:50:28 INFO SparkContext: SparkContext is stopping with exitCode 0 from stop at SparkFactorial.scala:18.
26/04/21 21:50:28 INFO SparkUI: Stopped Spark web UI at http://10.255.255.254:4040
26/04/21 21:50:28 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
26/04/21 21:50:28 INFO MemoryStore: MemoryStore cleared
26/04/21 21:50:28 INFO BlockManager: BlockManager stopped
26/04/21 21:50:28 INFO BlockManagerMaster: BlockManagerMaster stopped
26/04/21 21:50:28 INFO OutputCommitCoordinator\$OutputCommitCoordinatorEndpoint: OutputCommitCoordinator stopped!
26/04/21 21:50:28 INFO SparkContext: Successfully stopped SparkContext (Uptime: 5896 ms)
omkar@LAPTOP-S6G2S414:~/scala-basic\$